Georgia Institute of Technology

CS 4365: IEC

Spring 2026

Project Checkpoint Report

Project Name: Flight Delay Analysis and Performance Prediction

Names: Sishnukeshav Balamurali, Katherine Chang, Sanvi Thota

Project Plan (Plan):

This project focuses on the analysis of historical flight data and creating a predictive model on future performance of airlines through Machine Learning. In the past year, thousands of delays and cancellations have occurred due to weather events, staffing shortages, operational disruptions, and air traffic control constraints. These disruptions significantly impact passengers, airline operations, and airport logistics. Through data analysis, our team will visualize areas of concern and factors for these delays and use it to predict the status of upcoming flights.

Our goal is to create a data pipeline that covers:

(1) Data Collection - Use historical flight performance datasets from Kaggle and potentially supplement with APIs (Google Flights API or aviation data sources). We aim to collect one full year of 2025 data to capture seasonal variation

(2) Data Preprocessing - Clean missing values and normalize categorical features (airlines, airports), delay types, time-based features, and target variables

(3) Database Management - Store cleaned datasets in MySQL or SQLite, with potential AWS deployment. We plan to structure tables for flights, airlines, airports, and delay categories for more efficient querying

At the end of the semester, we plan to have a complete data pipeline that will connect to a notebook where our team will extract our clean dataset to create visualizations and ingest into ML models for predictions. Our **final deliverables** will include the cleaned dataset, a structured database, visual analysis of delay patterns, trained ML models for delay classification and prediction, and a final report summarizing our results.

Milestone chart:

Week	Sishnukeshav	Katherine	Sanvi
Week 1: 02/16	**Project Ideation**		
Week 2: 02/23	Data Collection	Data Cleaning And Preprocessing	Data Cleaning and Preprocessing
Week 3: 03/02	Database Setup	Data Cleaning and Preprocessing	Data Cleaning and Preprocessing
Week 4: 03/09	Visualizations	Visualization	Visualization
Week 5: 03/16	Database Query Optimization	ML: Architecture & Setup Planning	ML: Architecture & Setup Planning
Week 6: 03/23	**Spring Break**		
Week 7: 03/30	ML Model: Logistic Regression	ML Model: Random Forest	ML Models: Evaluation
Week 8: 04/06	Dashboard Development	Pipeline Integration	Dashboard Development
Week 9: 04/13	Video Prep/Recording	Presentation slides	Presentation slides
Week 10: 04/20	Documentation/Self-Evaluation	Documentation/Self-Evaluation	Documentation/Self-Evaluation

Project Requirements

- Data Collection: Kaggle, API (google flights), Open-source datasets
- Data Preprocessing: Python
- Database Management: MySQL/SQLite, AWS
- Notebook: Jupyter Notebook or Google Colab
- Machine Learning: predict flight delays/performance
- Data Visualization: pandas, matplotlib, seaborn (visuals)
- Dashboard/Presentation: Tableau/Power BI

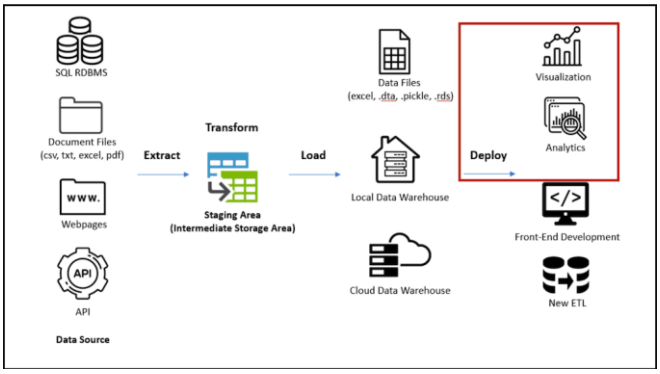

Figure 1. *ETL Pipeline layout example*

Possible Indicators for Delays:

I. Time of day
II. Location
III. Season
IV. Type of airline
V. Type of delay
VI. Shortage of workers at airports

Current progress report (Match):

Over the past two weeks, our team conducted project ideation, drafted the project proposal, and established our GitHub repository with an organized folder structure. We also identified our primary data source, the 2025 U.S. flight performance dataset available through Kaggle, and outlined our full data pipeline from collection through modeling. This work aligns with our planned Week 1-2 milestones as outlined in the chart above.

Over the next two weeks, we will begin active data collection, perform an initial exploration of the dataset to understand its structure and quality, and begin setting up our database

environment. No changes to the original plan have been made at this time, and the project remains on schedule.

Supporting Evidence (Factual):

We are currently looking into open-access datasets on Kaggle for maximum data accuracy.

URL to the GitHub project: https://github.com/katc1213/flight-performance-analysis
Dataset link:
https://www.transtats.bts.gov/DL_SelectFields.aspx?gnoyr_VQ=FGJ&QO_fu146_anzr=b0-gvzr

Skill Learning Report :

I. *ETL Data Pipeline (Data collection, cleaning, preprocessing)*
   Throughout the project phases, we will develop a data pipeline to automate the extraction of flight data from various sources, ensuring high data integrity and consistency. At this stage, we are gaining proficiency in handling large raw datasets and standardizing formats to prepare downstream analysis.
II. *Data Analysis*
   At this stage, we are developing skills in EDA using pandas, matplotlib, and seaborn. Throughout this process, we are improving our ability to interpret data patterns and identify meaningful relationships between variables, as well as communicate our insights clearly through visualizations.
III. *Relational Database Design*
   At this stage our team has begun reviewing MySQL and SQLite documentation to understand how to structure flight data across flights, airlines, airports, and delay categories.
IV. *Machine Learning Analysis*
   We have begun reviewing scikit documentation on logistic regression and random forest classifiers, and studying evaluation metrics including precision, recall, and F1 score.

Self-Evaluation:

I. *Scope (110/120): Our project spans a full data-science cycle including data collection, preprocessing, relational database design, exploratory analysis,*

visualization, and ML classification. Combined with the real-world complexity of flight delay data, and the breadth of tools and techniques involved, this project would be placed above a standard analysis project.

II. *Match (100/120): Our progress over the past two weeks aligns with Week 1 milestones. Specifically, project ideation, proposal drafting, and Github repository setup have all been completed. We are on track to begin data collection in the coming weeks with little to no deviations from the original timeline.*

III. *Factual (95/100): Our Github repository is live and organized with clearly structured directories, and a README. As data collection and preprocessing have not begun, we are not yet able to provide dataset samples & code execution results, which we expect to have available at the next checkpoint.*